**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 219 · Dijkstra: shortest paths II

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What vertex does Dijkstra settle next?

Q2. What does a weighted edge record?

Q3. What does a predecessor in a path mean?

Q4. Distance to A=3, edge A–B=4, current distance B=10. Update B.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

predecessor table — For each vertex, stores the prior vertex on its best-known path.

relaxation — Replace a distance when a cheaper route is discovered.

settled — Shortest distance is final for a vertex under non-negative edge weights.

infinity — Marker for an as-yet unreachable vertex.

route reconstruction — Follow predecessor links backwards then reverse.

## New learning

Track improvements — Relax each edge by comparing the candidate total with the current best-known distance.

Remember predecessors — When a distance improves, store the vertex from which the improvement came.

Finish with the best distance — Here E reaches distance 7 after A, C, B and D have settled.

Reconstruct backwards — Follow E←D←B←C←A, then reverse it to get the forward route.

Handle unreachable vertices — A vertex remaining at infinity cannot be reached from this start along the graph's edges.

## Worked example

Find the shortest route S→T and reconstruct it.

1. Predict the next state — Start S=0: tentative A=2, B=6.

2. Update the table — Settle A=2; B improves to 4 and T receives 9.

3. Update the table — Settle B=4; T improves to 5.

4. Update the table — Trace T←B←A←S, then reverse: S→A→B→T at cost 5.

# We Do · Practise with guidance

Find the shortest route P→T and reconstruct the predecessors.

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Edges: P-Q=4, P-R=1, R-Q=2, Q-T=3, R-T=10.
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Predict the next state

Start P=0 gives Q=4, R=1; settle R next.

Update the table

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.



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R improves Q to 3 and gives T=11.

Q improves T to 6.

T←Q←R←P reverses to P→R→Q→T, cost 6.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Starting at U, edges U–V=4, U–W=1 and W–V=2. Find the shortest U→V path and total cost.

### Question 2

Edges S–A=5, S–B=1, B–A=2, A–T=3 and B–T=9. Find S→T shortest distance, route and predecessor of T.

### Question 3

A graph contains a disconnected vertex Z. Explain what happens to tentative distance and predecessor for Z in Dijkstra, and when the algorithm may stop.

## Exit Ticket · Answer before checking

A predecessor table should be followed from the destination backwards before reversing.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. The unsettled vertex with the smallest current tentative distance.

Q2. The cost of travelling between two connected vertices.

Q3. The previous vertex on that route.

Q4. Candidate 3+4=7, so update B to 7 and predecessor(B)=A.

### You Do pages · Independent questions

Q1. [4] U→W costs 1 [1]; W→V adds 2 [1]; total 3 [1]; it is shorter than the direct cost 4 [1].

Q2. [6] Start B=1, A=5 [1]; settle B=1 [1], improve A=3 [1]; settle A=3, T improves to 6 [1]; route S→B→A→T [1]; predecessor(T)=A [1].

Q3. [6] Z is unreachable [1], remains at ∞ [1] and has no valid predecessor [1]; once all unsettled vertices have ∞ distance [1], no reachable vertex remains [1] and processing can terminate [1].

### Exit Ticket · Model answer

True. The predecessor of each node gives the step immediately before it on the path.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**